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**Term End Examinations – Fall Semester 2025-26 - December 2025**

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|---|---|
| Course code : EEE1001 | Slot :B14+B23+D21 |
| Course Title : **Electric Circuits and Systems** | Time: **180 Minutes** |
| Date/Session : **19th Jan, 2026** | Max. Marks: **100** |

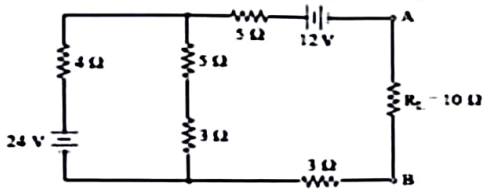
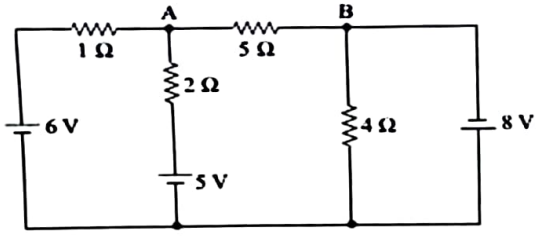
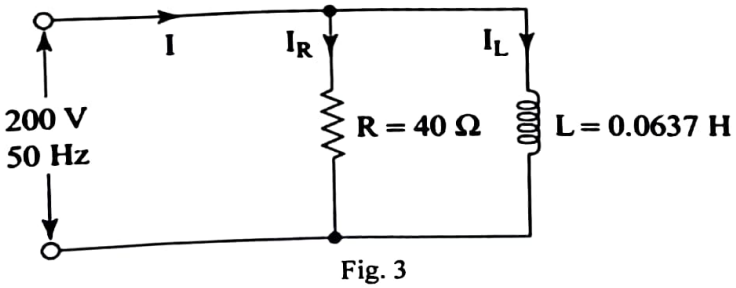
**KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE**

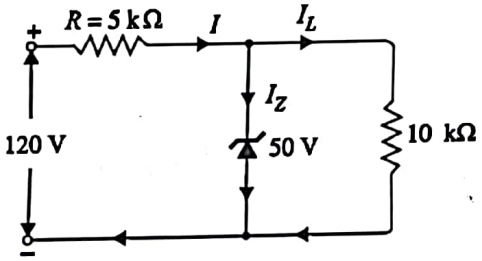
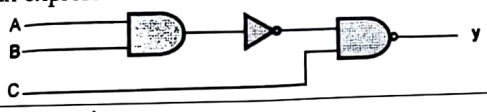
**General Instructions if any:**

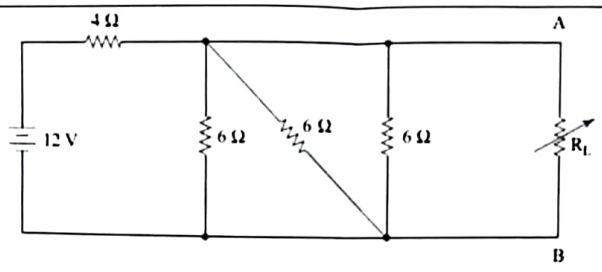
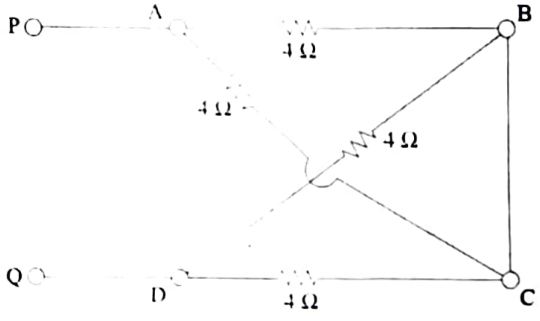
1. **Non-Programmable calculator is permitted: YES /NO**
2. **Reference tables are permitted: YES / NO (if Yes, Please specify:)**

**PART – A**

**Answer ANY FIVE Questions (5 X 12 = 60 Marks)**

| S. No. | | Questions | Marks |
|---|---|---|---|
| 1 | (a) | Using Thevenin's theorem calculate the current flowing through the load resistance RL connected across the terminals A and B as shown in Fig 1.   Fig. 1 | 12 |
| | | (OR) | |
| | (b) | For the circuit shown in Fig. 2, calculate the current flowing through the 5 Ω resistor by using the nodal method.   Fig. 2 | 12 |
| 2 | (a) | An alternating voltage of RMS value 100 V, 50 Hz is applied separately across a resistance of 10 Ω, an inductor of 100 mH, and a capacitor of 100 μF. Calculate the current flow in each case. Also draw and explain the phasor diagrams. | 7 |
| | (b) | In Fig. 3 is shown a parallel circuit, an inductance L and a parallel R connected across 200 V, 50 Hz ac supply. Calculate: (a) the current drawn from the supply; (b) apparent power; (c) real power; (d) reactive power.   Fig. 3 | 5 |

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| | | (OR) | |
| | (c) | A resistance of 50Ω, inductance of 29.8mH, capacitance of 3.4μF capacitor are connected in series across a 200V, 250HZ AC supply. Find (a) impedance of circuit (b) current (c) power consumed in the circuit (d) power factor (e) voltage drop across resistance (f) voltage drop across inductance (g) voltage drop across capacitance. Also draw the phasor diagram for the circuit. | 12 |
| 3 | (a) | Explain the construction and working principle of single-phase Transformer. | 6 |
| | (b) | The emf per turn of a single phase 10kVA, 2200/230V, 50 Hz transformer is 20V. Calculate (i)the number of primary and secondary turns.<br>(ii)the net cross-sectional area of core for a maximum flux density of 2T | 6 |
| | | (OR) | |
| | (c) | A magnetic circuit has a core of mean length 50 cm, cross-sectional area 4 $cm^2$, and permeability of $5\times10^{-3}$ H/m. If the coil wound around the core has 200 turns and carries a current of 2A, calculate the magnetic flux produced in the core. | 12 |
| 4 | (a) | Draw the circuit of a half-wave rectifier using a transformer and diode. Explain the function of each component with waveforms.<br>A half wave rectifier is fed from a voltage source of 66.64 V, 50 Hz. It is supplying a resistive load of 15 Ω. What is the average load current? | 12 |
| | | (OR) | |
| | (b) | Write short notes on Zener Diode.<br>For the circuit shown in Fig. 4, determine (i) the output voltage (ii) the voltage drops across series resistance (iii) the current through Zener diode.<br>  <br><br>Fig. 4 | 12 |
| 5 | (a) | Explain a half-subtractor and a half adder with a neat sketch of the circuit diagram, truth table, and its Boolean expression. | 12 |
| | | (OR) | |
| | (b) | Write the Boolean expression and Truth Table of the following Logic circuit.<br>  <br> | 6 |
| | (c) | Do the following conversions.<br>(i)$(1100101.0110)_2 = (?)_{10}$<br>(ii)$(415.214)_8 = (?)_2$<br>(iii)$(FE3.6D5)_{16} = (?)_{10}$<br>(iv)$(11110001.1101101)_2 = (?)_{16}$ | 6 |
| 6 | (a) | Calculate the value of load resistance $R_L$ for which maximum power transfer will occur from source to load. Also calculate the value of maximum power. | 12 |

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|---|---|---|---|
| | |  Fig. 5 | |
| | | (OR) | |
| | (b) | Discuss the limitations of the analogy between electric circuits and magnetic circuits. Magnetic series and parallel circuits behave in a manner similar to series and parallel electrical circuits." Discuss this statement in detail. | 12 |
| | | **PART – B** | |
| | | **Answer ANY FIVE Questions (40 Marks)** | |
| 7 | | Compute the resistance between the terminals P and Q of the network shown.  | 8 |
| 8 | | A capacitor of value 1μF and a resistor of 5.45 MΩ are connected in series across a 220 V dc supply through a switch. Calculate the time by which the capacitor will be charged to 60% of the supply voltage. | 8 |
| 9 | | Briefly explain the construction and working principle of a three-phase induction motor with the help of a neat diagram. | 8 |
| 10 | | Illustrate the working of p-n-p transistor with neat diagram and explain its input and output characteristics. | 8 |
| 11 | | (a)Draw and explain the internal circuit diagram of a 4:1 multiplexer.<br>(b)Design S-R Flip Flop using NAND gate and write its truth table. | 4+4 |
| 12 | | Describe the operation of a PN junction under forward bias and reverse bias. | 8 |

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